

Condition Monitoring Strategy

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

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**August 2007
1007-2.0-1102**

1.0 Introduction

The demands for increased production and reduced cost per ton in the mining industry have caused mining companies to look for new ways to increase equipment availability and reduce operating costs. Equipment maintenance is a key contributor to overall operating costs of production loading and haulage equipment. It has been demonstrated that the total costs associated with unscheduled downtime events (failures) can be as much as fifteen times that of the same repair performed on a scheduled basis when factors such as reduced parts reuse due to contingent damage, incremental labor (inefficient use of manpower), and lost production as a result of increased equipment repair downtime are considered.

Availability is driven by the frequency and duration of machine downtime events (failures and preemptive shutdowns). Studies have shown that shutdown frequency is the primary element in this inter-dependency. Shutdown frequency is a function of inherent machine failure rate and equipment maintenance management effectiveness. Traditional statistical machine reliability calculations are based on failure rates; they ignore preemptive maintenance shutdowns which occur as a result of preventive and predictive (non-failure-related) actions which are focused on problem / failure avoidance.

As such, many mining companies have adopted a repair-before-failure strategy in the interest of reducing the number of unscheduled downtime events. A well-conceived, fully functional Condition Monitoring plan to support the timely and accurate detection of changes in equipment health / performance and operation / application severity is the foundation of any condition-based and repair-before-failure maintenance strategy.

This Best Practice relates to **version II of Site Assessment items 2.13.1 - 2.13.8.**

2.0 Best Practice Description

The first step in failure avoidance is the awareness that potential problems or conditions that may lead to potential machine shutdowns exist. Condition Monitoring is the means by which we detect those problems and associated machine condition. Condition Monitoring is based on proactive data acquisition and analysis routines (processes & sub-processes) used together to facilitate early detection of machine defects or changes in machine health and application severity enabling proactive maintenance, repair, and application management to take place. Optimal Condition Monitoring routines for mobile equipment include:

- ✓ Operator Inspections,
- ✓ Periodic / Daily Inspections,
- ✓ pre-PM Inspections,
- ✓ PM Inspections,
- ✓ post-PM Audit Inspections,
- ✓ pre-PCR Inspections,
- ✓ Fluids Management & Analyses,
- ✓ On-board Machine Electronic Data Monitoring,
- ✓ Machine Systems Performance Tests,
- ✓ Application Analyses, and
- ✓ Learning from Failures / Failure & Wear Analyses.

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The following is a brief description of each of those Condition Monitoring routines:

- Operator Inspections

Operators spend more time in the machine, and as a result are more familiar with the equipment and its operation than anyone involved in its management. The scope of the operators' involvement in Condition Monitoring includes pre-shift inspection, machine condition & performance evaluation during the shift, and post-shift operational assessment. Roles and responsibilities of the operators in these activities include:

- pre-Shift Inspection
 - ✓ Review machine log book; discuss with Operator from previous shift,
 - ✓ Perform visual, walk-around inspection,
 - ✓ Record findings ... pre-shift checklist & machine log book,
 - ✓ Report defects to appropriate area (Dispatch / in-pit supervision).
- Machine Condition & Performance Evaluation during the Shift
 - ✓ Monitor machine condition & performance as ongoing process,
 - ✓ Use all the "tools" at his disposal,
 - ✓ Use / activate on-board monitoring system(s),
 - ✓ Report defects / abnormalities to appropriate area (Dispatch / in-pit supervision),
 - ✓ Learn from "failures".
- post-Shift Operational Assessment
 - ✓ Perform post-shift machine performance assessment,
 - ✓ Record observations ... post-shift checklist & machine log book,
 - ✓ Discuss with Operator of next shift.

Characteristics of the pre-shift and post-shift inspection checklists are:

- pre-Shift Inspection Checklist
 - ✓ Designed to capture visual observations,
 - ✓ Low tech; no tools or instrumentation needed,
 - ✓ Brief (<10 minutes); easy to complete,
 - ✓ Minimum training required.
- post-Shift Inspection Checklist
 - ✓ Designed to capture subjective operational abnormalities,
 - ✓ Low tech ... "seat of the pants",
 - ✓ No tools or instrumentation needed,
 - ✓ Brief (<5 minutes); easy to complete,
 - ✓ Minimum training required.

- Periodic / Daily Inspections

Daily inspections should be scheduled during convenient "windows of opportunity" such as the daily lubrication and refuel stop. These inspections should be performed by a highly skilled, detail-oriented Inspector, are typically quick and easy to perform (15 to 20 minutes duration), and tend to be based on low-tech, visual, subjective observations although items such as tire pressure checks and fluid additions are frequently included.

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The daily inspection also includes:

- an Operator interview,
- a review of available information (Operators' logbook, backlog repair list, top problems summary, etc.),
- and a visual inspection.

Critical to the success of the daily inspection are well-guided, machine-specific inspection checklists, well-documented procedures, minimal reliance on verbal communications, and the Inspectors' ability to analyze, interpret and translate Operator input into technical feedback and action such as backlog repair requests.

• pre-PM Inspections

The pre-PM inspection is performed to verify that the list of activities planned for Preventive Maintenance is accurate and complete. The pre-PM inspection must be scheduled and executed in time to allow Planning to respond to feedback from the inspection. Pre-PM inspections are typically allow enough time for the fluid lab to come back with the associated results and interpretation.

Pre-PM inspections) are the responsibility of the Field Inspectors and Technicians. In addition to all of the activities normally done during the daily inspection activities such as fluid sampling, magnetic plug inspections, VIMS & other on-board system downloads, and backlog repair validation are included in the pre-PM inspection. The pre-PM inspection should be accomplished during the same "windows of opportunity" as the daily inspection and in the same amount of time (15 to 20 minutes duration). The same factors that drive success of the daily inspection are also critical to the success of the pre-PM inspection.

• PM Inspections

One of most important goals and objectives of Preventive Maintenance is efficient, effective execution of periodic preventive and predictive maintenance routines. Since Preventive Maintenance is performed on a regular basis (typically every 250 to 500 hours) and the times allocated to PM can be somewhat lengthy (a shift or more), it is possible to execute several Condition Monitoring routines during the "Window of Opportunity" presented by PM. These include Operator interviews, fluids management, on-board system checks, and machine systems performance tests, all of which are described and discussed separately in this document.

In addition to those mentioned in the previous paragraph, the following Condition Monitoring activities are normally associated with and executed during Preventive Maintenance:

- pre-Wash Inspection ... inspect before the evidence is washed away.
 - ✓ Perform visual, walk-around inspection for ...
 - Frame & structural cracks, weld failures, etc.,
 - Loose, missing, broken bolts fasteners, clamps, etc.,
 - Oil / fuel/ coolant / air leaks,
 - Accident damage, e.g. bent sheet metal, broken items, etc.
 - ✓ System performance tests on ...
 - Air system,

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- Air compressor,
 - Service brakes,
 - Parking brake,
 - Engine,
 - Hoist circuit,
 - Traction control system.
- Preliminary PM Inspection ... tasks better performed outside the PM bay.
 - ✓ Inspect for wear or excessive movement of:
 - Hoist cylinder mounts (pins & bearings),
 - Steering linkage (ballstuds, pins & bearings),
 - Steering system cycle time,
 - A-frame bearing wear,
 - Body pivot pin & bushing,
 - Body safety cable,
 - Suspension cylinder pressures, ...
 - PM Inspection ... tasks performed inside the PM bay.
 - ✓ Inspect for leaks, loose or missing fasteners, lines / hose routing, cracks and other damage in the following areas:
 - Engine compartment (above, inside & below),
 - Front corners (brakes, struts, steering linkage, ...),
 - Center of machine (above& below),
 - Rear of machine (above& below),
 - Access / egress systems,
 - Operator's station,
 - Upper platform,
 - Frame.
 - post-PM Audit Inspections

Executing the Preventive Maintenance program as planned can mean the difference between success & failure. Auditing the PM process is the best way to ensure compliance with the PM plan. Post-PM Audit Inspections can be used to enhance PM quality and effectiveness by defining PM training opportunities, refining the PM checklists, and identifying shortcomings in the PM process. If properly communicated, the audit results can be used in an ongoing continuous improvement effort to refine the PM process to meet site-specific requirements. The PM audit process can be accomplished in several ways, Supervisor sign-off at critical points in the process (preferred method) and random checks after PM completion being the two most common approaches.
 - pre-PCR Inspections

The pre-PCR Inspection is a comprehensive assessment of machine condition performed prior to planned major component replacement. The intention is to identify all of the defects that exist on a piece of equipment far enough in advance that preventive and corrective actions can be taken in conjunction with the “window of opportunity” presented by the PCR event with the ultimate goal of returning the machine back into service with zero defects after the PCR. Engine

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replacement usually provides the best opportunity since downtime duration associated with engine replacement is greater than most other components.

Participants in this inspection include representatives from the Field Inspection team, Shop Staff and Supervision, the Parts Department and the Fleet Analyst. Pre-PCR inspection activities include:

- fluid sampling,
- magnetic plug inspection,
- on-board electronic system downloads,
- detailed visual inspections,
- special instrumented testing, and
- backlogged repair validation.

The pre-PCR Inspection team uses detailed, machine-specific checklists developed from the repair / downtime history, the pending backlog list and the top problem summary for the machine in question as key inputs that guide their work.

The pre-PCR Inspection should require no more than one shift (typically 4 to 8 hours) to perform. It should be scheduled and executed to permit Planning response to the pre-PCR results. Once the pre-PCR Inspection is completed, a machine-specific plan should be developed and scheduled for execution during the PCR. The Fleet Analyst should establish goals for improvement, e.g. availability and MTBS, and monitor performance after the PCR to determine if those goals have been achieved.

• Fluids Management & Analyses

This sub-process involves multiple routines including:

- scheduled oil sampling (SOS),
- particle count,
- “kidney looping” (fluids dialyses),
- magnetic plug inspections,
- filter / breather / strainer / screen inspections, and
- fluid consumption, e.g. oil, fuel, coolant, grease.

The objectives of fluid management and analyses are:

- assess equipment condition in terms of :
 - o compartment cleanliness,
 - o component wear and fluid leakage,
 - o transfer and/or consumption, e.g. engine oil burn.
- Assess oil condition in terms of:
 - o Oil cleanliness (ISO count),
 - o Oil condition (oxidation, additive contents, etc.)

In addition, fuel burn rate can be used as an indication of machine application severity. Due to the importance of fluid cleanliness and contamination control, it is important to include monitoring of bulk storage and fluid handling / distribution in the Condition Monitoring strategy.

• On-board Machine Electronic Data Monitoring

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Machine electronics have evolved in the area of systems control. This has opened many doors for electronics to monitor equipment performance, health, operating techniques and application severity. VIMS, ECM's, and other electronic monitoring and control systems contain a wealth of information that can be used to detect potential failures and identify operational practices or application characteristics that dictate changes to the operation / application or modifications to the maintenance strategy designed to deal with those changes.

In order for the information to be useful, the Condition Monitoring strategy needs to clearly define procedures for diligent data collection. The data then needs to flow into a well-organized system where it can be:

- trended,
- compared to pre-defined performance targets or application criteria,
- analyzed and,
- corrective action(s) taken.

• Machine Systems Performance Tests

Machine systems tests are used to confirm proper performance or to monitor and trend degradation in performance relative to baseline (commissioning) test results and factory specifications defining acceptability criteria. They typically require some form of special tooling or instrumentation and are primarily focused on major components and systems, e.g. engine, transmission, torque converter, steering, brakes, or hydraulics. These tests are designed to measure specific system and component performance including both static and dynamic checks, e.g. system speed, temperature and pressure tests, torque stall, brake hold performance, etc. Machine systems tests follow specific checklists and procedural guidelines and are typically performed during PM and after repair(s) to ensure that proper system / component performance has been restored.

• Application Analyses

The application environment in which a piece of equipment is used in has a direct impact on the overall performance of that equipment. Given the fact that changes in machine application with respect to haul road grades (grades become steeper or shallower), time spent on grade (increasing pit depth), haul distances (hauls typically become longer), etc., it would be expected to see variations in application severity over time.

Maintenance must recognize the impact of changes in application severity and modify maintenance practices and the maintenance management strategy accordingly, e.g. the component replacement plan.

Ongoing evaluation of mine operations via tracking, trending and reporting application severity should be a standard practice of any Maintenance Department. Proactive analysis and interpretation of those results is the first indication to management to initiate a more in-depth investigation, e.g. modification of operational practices or of the maintenance strategy. The reactive alternative is to wait until the equipment "tells" management that something in the application has changed through premature component failures.

Application characteristics are critical variables that must be considered in the definition of and subsequent revisions to the maintenance strategy, particularly with respect to component life

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targets. Application severity analysis routines are intended to contribute to an overall understanding of the Condition Monitoring picture via the short, medium, and long-term assessments of mine severity.

Data sources are typically on-board electronic systems (primarily VIMS or TPMS) and mine production and dispatch reports. In a haul truck scenario the most critical application parameters include:

- fuel burn rate,
- haul cycle detail, including:
 - o haul cycle distances
 - o cycle times and
 - o number of haul cycles
 - o travel speeds for each segment in the overall cycle,
- payload management,
- haul road conditions,
- production,
- transmission shift count / frequency,
- etc.

Equipment-specific guidelines and acceptability criteria for each machine must be identified to quantify the impact of application on performance.

• Learning from Failures / Failure & Wear Analyses

The Condition Monitoring process include analysis routines for all failures and unscheduled downtime events with the clear goal of detecting the root cause of the failure / event and a determination as to why the operation was unable to prevent failures from happening and/or what could be done differently to prevent them from happening again. In establishing these routines it will be needed to consider input from each of the following:

- Analysis of machine repair history:
 - ✓ Pareto Analysis
 - ✓ Top Ten ranking by MTBS, MTTR, impact on availability & costs
- Failure Analysis:
 - ✓ Root cause failure analysis of failures & unscheduled downtime events
- Analysis of effectiveness of:
 - ✓ Repair actions
 - ✓ Detection routines
- Input from other mines with:
 - ✓ Similar equipment
 - ✓ Similar application
- Input from the manufacturer:
 - ✓ Technical Information Bulletins
 - ✓ Service Magazine articles
 - ✓ Factory programs

The Condition Monitoring strategy should define the linkage between the failure analysis and the steps required to implement changes via the Continuous Improvement process.

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3.0 Implementation Steps

Critical steps associated with the implementation of an on-site Condition Monitoring process include the definition and development of:

- Condition Monitoring Strategy
 - ✓ Contract scope, goals & commitments
 - ✓ Fleet characteristics
 - ✓ Condition Monitoring routines
 - ✓ Manpower requirements
 - ✓ Training requirements
 - ✓ Resource requirements
 - ✓ Performance Metrics
 - ✓ Analysis and Interpretation
- Plan Definition
 - ✓ Which Condition Monitoring routines will be required to implement the strategy, i.e. Operator Inspections, Periodic / Daily Inspections, pre-PM Inspections, ... ?
- Manpower
 - ✓ What human resources are required to implement and execute the Condition Monitoring plan, i.e. roles & responsibilities, personal attributes, quantity, skills, ... ?
- Training
 - ✓ How can we assess personnel competency? What training will be required to develop the skills required to perform the tasks related to Condition Monitoring?
- Resources
 - ✓ What resources are required to execute the Condition Monitoring plan, i.e. machine-specific inspection checklists, well-defined procedures, special tooling, instrumentation, dedicated equipment, data management system, external resources, ... ?
- Performance Metrics
 - ✓ What defines success in Condition Monitoring? Which performance measures are available that aid in quantifying our efforts in Condition Monitoring? Once these questions have been answered, the following issues need to be taken into account when developing a plan to quantify Condition Monitoring performance:
 - Raw input data requirements
 - Data sources
 - Data collection frequency & responsibilities
 - Data flow & management
 - Calculation methodology

Please see the document “Metrics (KPI’s) to Assess Process Performance” for a complete listing and explanation of Condition Monitoring process performance metrics.

- Analysis and Interpretation

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- ✓ Data are neither the goal nor the end result; it is the means by which we gain an understanding of what is occurring. The analysis and interpretation routines are sub-process by which all Condition Monitoring results are collectively analyzed and interpreted taking into consideration all possible relationships and potential causes. These routines help guide us to the best possible conclusions and subsequently to the best recommendations and corrective action plans. The following issues need to be considered when developing the plan to analyze and interpret Condition Monitoring results:
 - Performance goals & acceptability criteria
 - Analysis & interpretation roles & responsibilities
 - Analysis & interpretation guidelines
 - Information management / archiving
 - Report formats (templates)
 - Report timing & frequency
 - Communications / distribution network
 - Distribution format & media
 - Linkage to Production / Operations
 - Linkage to the Continuous Improvement process

4.0 Benefits

Successful implementation and execution of the Condition Monitoring process will:

- facilitate timely and accurate detection of changes in:
 - o equipment health
 - o performance and operation
 - o application severity
- reduce the number of failures and unscheduled downtime repair events
- favorable impact on overall operation and maintenance costs,
- efficient use of labor resources,
- equipment reliability / availability
- improving production and cost per ton.

5.0 Resource Requirements

- Human Resources:

Human resource requirements associated with the execution of the Condition Monitoring plan include:

- ✓ Fleet Analyst (Reliability Engineer) ... provides leadership and direction for all on-site Condition Monitoring activities.
 - Define Condition Monitoring routines that correlate to equipment health, operation & application.
 - Review / revise Condition Monitoring procedures & checklists to ensure they focus on significant issues.
 - Ensure accuracy & completeness of data acquired through Condition Monitoring.
 - Evaluate / interpret all Condition Monitoring data.
 - Advise Planning of issues related to equipment health, maintenance, application & operation.

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- ✓ Planner ... develops plans including resource & time estimates to execute Condition Monitoring routines.
 - Ensure that Condition Monitoring activities are aligned with the strategy.
 - Define, organize & procure facilities, manpower, parts, tooling, forms, checklists, etc. for Condition Monitoring.
 - Establish priorities for Condition Monitoring execution.
 - Monitor maintenance and repair performance metrics for compliance with goals; devise & communicate plans to address shortfalls to the Project Manager.
 - Monitor Condition Monitoring performance; correct as necessary.
- ✓ Scheduler ... translates the Condition Monitoring plan into an actionable schedule.
 - Identifies “Windows of Opportunity” for execution of Condition Monitoring activities.
 - Schedules weekly Condition Monitoring activities in accordance with the Condition Monitoring plan & strategy and resource availability.
 - Publishes / distributes the Condition Monitoring schedule & obtains agreement from all appropriate areas.
 - Adjusts the Condition Monitoring schedule to comply with variations in equipment utilization & resource availability.
- ✓ Operators ... identify defects & equipment health issues on an ongoing basis before during & after the shift.
 - Continuously assess machine condition, health & performance.
 - Document results of his assessment through pre & post-shift checklists and the machine logbook.
 - Take appropriate preventive action in response to input from on-board warning systems.
 - Communicate machine condition & performance issues to Dispatch, Field Service & Inspectors.
- ✓ Field Inspectors ... participates in all Condition Monitoring activities in the field in accordance with the CM strategy / plan.
 - Translate Operator input into root-cause technical information.
 - Execute all periodic / daily field inspections.
 - Participate in the execution of pre-PM inspections.
 - Participate in the execution of pre-PCR inspections.
 - Provide accurate, complete information on inspection checklists and field documentation.
 - Assess priority of defects detected via field Condition Monitoring tasks.
 - Generate / enter backlog repair requests.
- ✓ Technicians ... perform pre-PM, PM & pre-PCR Condition Monitoring activities in accordance with the Condition Monitoring strategy / plan.
 - Become familiar with proper use & operation of all Condition Monitoring tooling & instrumentation related to pre-PM, PM & pre-PCR activities.

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- Adhere to guidelines & procedures defined for those Condition Monitoring activities.
 - Document results; provide accurate, complete information on checklists & forms.
 - Generate backlog repair requests for defects that cannot be addressed immediately.
- ✓ Project Manager ... primary on-site individual responsible for the implementation and execution of the Condition Monitoring strategy.
- Total familiarity with customer expectations, contractual commitments and the Condition Monitoring strategy.
 - Authorizes resources to support successful implementation of the Condition Monitoring strategy.
 - Audit fleet and Condition Monitoring process metrics; takes action to eliminate any deficiencies.
 - Authorizes revisions to Condition Monitoring strategy and procedures.
 - Authorizes modifications to the component exchange program based on Condition Monitoring input.
- Logistical Resources:
- Logistical resource requirements associated with the execution of the Condition Monitoring plan include:
- ✓ Facilities
- Designate specific locations for each Condition Monitoring activity ... repair shop bays, PM bays, fuel island / bay ...
 - Allocate office space for Condition Monitoring data collection and analysis.
- ✓ Tooling
- Dedicated special tooling ... filter cutting device, oil extraction (SOS), kidney looping, etc.
 - Dedicated diagnostic instrumentation ... pressure / temperature / speed gauge packages, particle counter, etc.
 - Dedicated service / technical literature.
 - Personal safety equipment.
- ✓ Equipment
- Field transportation ... pick-ups for field use,
 - Communications devices ... to / from Dispatch, field, shop, Condition Monitoring office,
 - Computer(s) dedicated to information access, Condition Monitoring archive ...
 - Computerized maintenance management system ... software package for data management, information analysis and interpretation, and report generation.
- ✓ Information
- Weekly schedule for Condition Monitoring routines,

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- Work orders for Condition Monitoring & related Condition Monitoring activities, e.g. Preventive Maintenance,
 - Condition Monitoring checklists and procedures specific to each routine for every model / equipment,
 - Interpretation guidelines and acceptability criteria,
 - Special instructions, etc.,
 - Training (dealer Training Dept).
- ✓ External Resources (access to)
- Oil Lab (SOS / particle count),
 - CRC (wear measurements, failure analyses ...),
 - Service providers for wear measurements, vibration analyses, ...
 - Training (external training providers).

Significant implementation costs include logistical resources (primarily the required equipment and tooling), the time invested in material content development (checklists, procedures, guidelines, etc.), training for the Condition Monitoring staff, and the acquisition of personnel with the capabilities to execute the Condition Monitoring process.

6.0 Supporting Attachments

- Cat Global Mining Equipment Management Process Map (Condition Monitoring)
- “Metrics (KPI’s) to Assess Process Performance”  PDF Document

7.0 Related Best Practices

[0806-2.70-1007](#) [Condition Monitoring – Field Inspections](#)
[0806-2.70-1008](#) [Backlog Management](#)

8.0 Acknowledgements

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Special recognition to Daniel Adkison, Fred Ringness, and Abelardo Flores for their contributions.

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